

2 Supplemental Tables

Table S1. Unique EcM fungal sampling locations per Canadian ecozone (National Ecological Framework), for the combined GlobalFungi + GenBank dataset ($n = 1,623$ locations across 15 ecozones) and for GlobalFungi only ($n = 486$ locations), with ecozone area and combined sampling density. Points are assigned to ecozones by a within-polygon join against the detailed (unclipped) ecoregion layer, with points falling just offshore of the ecoregion coastline snapped to the nearest ecozone within 5 km; ecozone areas use the same polygon basis, so counts and areas are consistent. Ordered by decreasing combined density.

Ecozone	GlobalFungi (loc.)	Total (loc.)	Area (km ²)	Density (loc. / 10,000 km ²)
Mixedwood Plains	58	387	169258	22.86
Pacific Maritime	2	209	251087	8.32
Atlantic Maritime	104	156	282969	5.51
Boreal Shield	222	667	2079895	3.21
Montane Cordillera	17	60	489977	1.22
Taiga Plains	46	47	657947	0.71
Boreal Plains	20	38	740741	0.51
Boreal Cordillera	6	16	470758	0.34
Taiga Shield	9	24	1453302	0.17
Arctic Cordillera	1	2	340729	0.06
Northern Arctic	0	13	2838182	0.05
Prairies	1	2	466981	0.04
Southern Arctic	0	2	1683284	0.01
Hudson Plains	0	0	448432	0.00
Taiga Cordillera	0	0	266972	0.00

Table S2. The 28 FungalRoot/BIEN native Canadian EcM host species with at least one recorded, species-level host taxon in Canadian GlobalFungi or GenBank EcM records (the intersection between the native host list and the distinct species-level host taxa recorded across both sources).

Host species
<i>Abies balsamea</i>
<i>Alnus alnobetula</i>
<i>Alnus incana</i>
<i>Betula nana</i>
<i>Betula papyrifera</i>
<i>Dryas integrifolia</i>
<i>Gaultheria shallon</i>
<i>Monotropa uniflora</i>
<i>Picea engelmannii</i>
<i>Picea glauca</i>
<i>Picea mariana</i>
<i>Picea rubens</i>
<i>Pinus albicaulis</i>
<i>Pinus banksiana</i>
<i>Pinus contorta</i>
<i>Pinus ponderosa</i>
<i>Pinus rigida</i>
<i>Pinus strobus</i>
<i>Populus balsamifera</i>
<i>Populus tremuloides</i>
<i>Populus trichocarpa</i>
<i>Pseudotsuga menziesii</i>
<i>Pterospora andromedea</i>
<i>Quercus garryana</i>
<i>Quercus rubra</i>
<i>Salix arctica</i>
<i>Salix reticulata</i>
<i>Tsuga heterophylla</i>

Table S3. The 22 species-level host names recorded in the Canadian GlobalFungi or GenBank metadata that did not match the FungalRoot/BIEN native EcM host species list (Table S2). These include ornamental or non-native species, misspellings, and other problematic or truncated entries, and are excluded from the host-interaction analyses.

Host name (metadata)
<i>Abies balsamifera</i>
<i>Alnus crispa</i>
<i>Ascophyllum nodosum</i>
<i>Betula in</i>
<i>Betula payrifera</i>
<i>Corylus avellana</i>
<i>Deschampsia cespitosa</i>
<i>Douglas fir</i>
<i>Fagus or</i>
<i>Fagus sylvatica</i>
<i>Kalmia angustifolia</i>
<i>Not available</i>
<i>Oak beech</i>
<i>Picea abies</i>
<i>Pinus mugo</i>
<i>Pinus radiata</i>
<i>Populus alba</i>
<i>Populus and</i>
<i>Populus in</i>
<i>Quercus robur</i>
<i>Ulva intestinalis</i>
<i>Zostera marina</i>

Table S4. The 32 EcM fungal genera (FungalTraits) represented in the Canadian GBIF physical-specimen records (Figure S5) that have no sequence data in Canada (not detected in the combined GlobalFungi + GenBank dataset), and so could be targeted for future molecular sampling. 'GBIF records' and 'GBIF species' are the count of physical-specimen occurrence records and distinct species names per genus in the GBIF download.

Genus	GBIF records	GBIF species
Boletinus	276	5
Turbinellus	190	2
Imleria	125	2
Gomphus	115	1
Albatrelloopsis	109	3
Xanthoconium	84	2
Wynnella	41	2
Hortiboletus	33	2
Boletellus	27	1
Laeticutis	21	1
Gyrodon	19	1
Aleurina	18	3
Baorangia	14	1
Tremellodendron	14	3
Phallogaster	12	1
Cyanoboletus	11	3
Neoboletus	11	3
Underwoodia	10	1
Gastroboletus	6	2
Paragyrodon	6	1
Bankera	5	2
Boletochaete	5	1
Pseudocraterellus	5	2
Riessia	4	1
Porpoloma	3	1
Hydnangium	2	1
Plicariella	2	2
Zelleromyces	2	2
Brauniellula	1	1
Marcelleina	1	1
Mycolevis	1	1
Protoglossum	1	1